

4. Motion in the Yukawa potential: A particle moves in the Yukawa potential described by

$$V(r) = -\frac{k}{r} \exp(-r/a),$$

where k and a are positive quantities.

- (a) Write down the equations of motion and reduce them to the equivalent one-dimensional problem.
- (b) Use the effective potential to discuss the qualitative nature of the orbits for different values of the energy and the angular momentum.
- (c) Also, show that if the orbit is nearly circular, the apsides will advance approximately by $\pi \rho^2/a^2$ per revolution, where ρ is the radius of the circular orbit.

Note: The radial distances to the turning points are known as apsidal distances.

b)

$$V_{\text{eff}}(r) = \frac{L^2}{2mr^2} - \frac{K}{r} e^{-\frac{r}{a}}$$

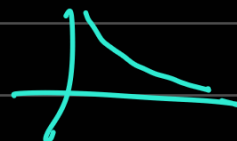
$$V_{\text{eff}}(r) = \frac{L^2}{2m} \left(\frac{1}{r^2} - \frac{2mK}{L^2} \frac{1}{r} e^{-\frac{r}{a}} \right)$$

let $A = \frac{L^2}{2m}$, $b = \frac{L^2}{2mK} \rightarrow$ dimensions of length

$$\Rightarrow V_{\text{eff}} = A \left(\frac{1}{r^2} - \frac{e^{-\frac{r}{a}}}{br} \right)$$

Observe: $V \rightarrow \infty$ as $r \rightarrow 0$

$V \rightarrow 0$ as $r \rightarrow \infty$

If b is large then $V \sim \frac{1}{r^2}$ 

But if b small enough, extrema possible.

Let's be more quantitative.

$$\begin{aligned} \frac{dV_{\text{eff}}}{dr} &= A \left(-\frac{2}{r^3} + \frac{e^{-\frac{r}{a}}}{br^2} + \frac{e^{-\frac{r}{a}}}{abr} \right) \\ &= \frac{A}{r^3} \left(\frac{r^2}{ab} e^{-\frac{r}{a}} + \frac{r}{b} e^{-\frac{r}{a}} - 2 \right) \end{aligned}$$

$$\frac{dV_{\text{eff}}}{dr} = \frac{aA}{br^3} \left(\frac{r^2}{a^2} e^{-\frac{r}{a}} + \frac{r}{a} e^{-\frac{r}{a}} - 2 \frac{b}{a} \right)$$

Circular orbits are possible when $\frac{dV_{\text{eff}}}{dr} = 0$

$$x = \frac{r}{a}$$



$$\Rightarrow x^2 e^{-x} + x e^{-x} = \frac{2b}{a}$$

— (1)

Now let's investigate $f(x) = e^{-x} x(1+x)$

What is its maximum value?

Verify for yourself
that it is maxima.

$$f'(x) = -e^{-x} x(1+x) + e^{-x}(1+x) + e^{-x} x = 0$$

$$-x(1+x) + (1+x) + x = 0$$

$$-x - x^2 + 1 + 2x = 0$$

$$\Rightarrow x^2 - x - 1 = 0$$

$$\Rightarrow x = \frac{1+\sqrt{5}}{2}$$

→ Golden Ratio!

$$\text{So at } x_0 = \frac{1+\sqrt{5}}{2},$$

$$f(x) = e^{-x_0} x_0^3 = \underline{0.84}$$

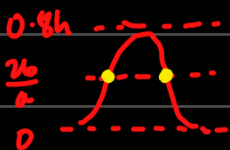
So from (1), $\rightarrow$ if $\frac{2b}{a} > 0.84$, then $\frac{dV_{\text{eff}}}{dr} < 0 \forall r > 0$

$\rightarrow$ if $\frac{2b}{a} = 0.84$ then $\frac{dV_{\text{eff}}}{dr} = 0$ for one r .

$\rightarrow$ if $\frac{2b}{a} < 0.84$ then $\frac{dV_{\text{eff}}}{dr} = 0$ for two r

This is because $f(x) = 0$ at $x=0$, goes up till
a maximum $= 0.84 > \frac{2b}{a}$ and ultimately
must go to $f(x) = 0$ at $x \rightarrow \infty$

So $f(x) - \frac{2b}{a}$ must be zero twice in $[0, \infty)$

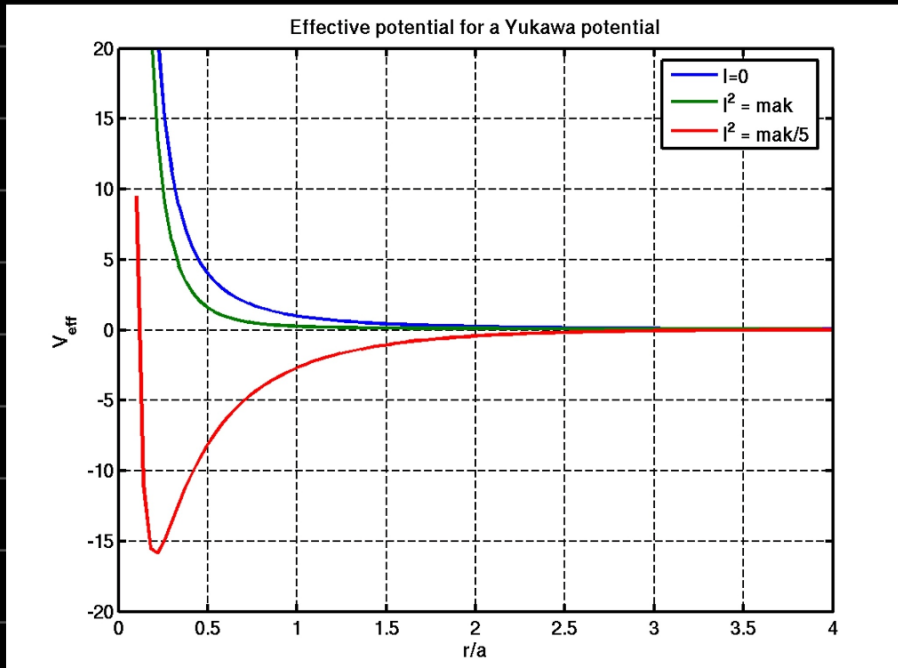


And so $\frac{dV_{\text{eff}}}{dr}$ must be zero twice.

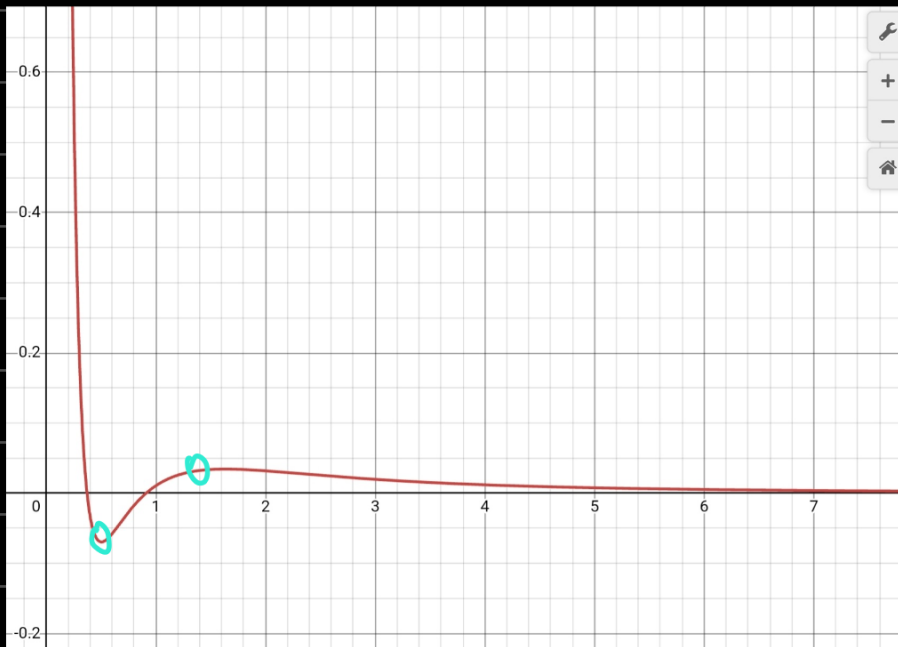
The critical angular momentum l_c is then given by

$$\frac{2b}{a} = 0.84$$

$\Rightarrow l_c^2 = 0.84 \text{ amk}$, after which circular orbits cannot exist.



Graphs for three different $\frac{2b}{a}$ values.

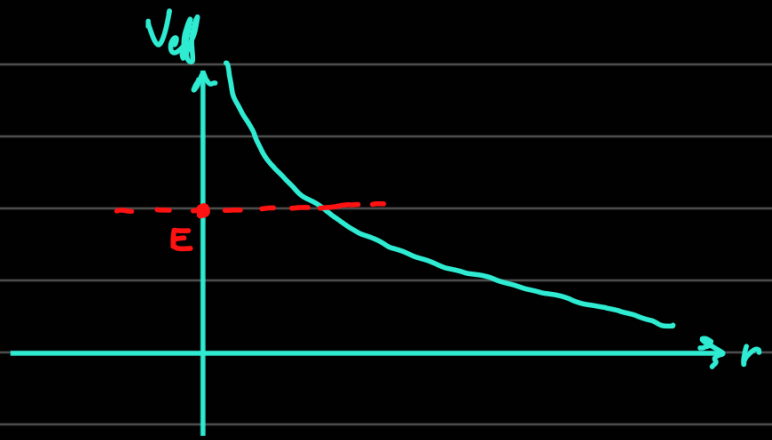


Notice two extrema !

Lastly, let's discuss how choosing different E affects our solution.

Case 1 : $\frac{2b}{a} > 0.84$

Any choice of E
leads to the particle
repelled away from
origin.

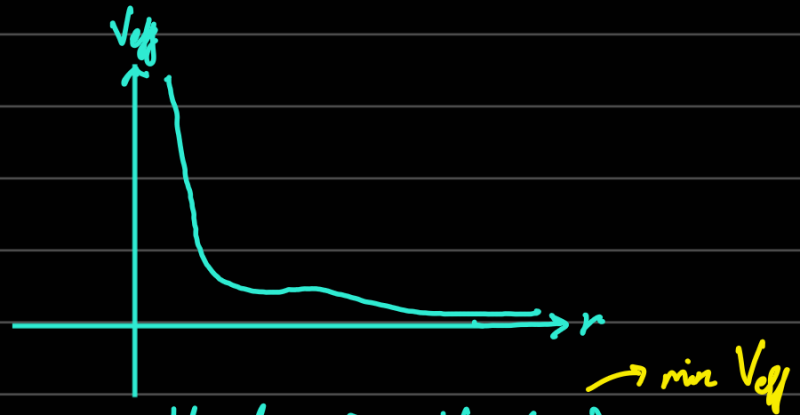


Case 2 : $\frac{2b}{a} = 0.84$

$\frac{dV_{eff}}{dr} < 0$ always, but vanishes
for $f(x_0) = 0.84$

So $x_0 = \frac{1+\sqrt{5}}{2}$ is an inflexion point.

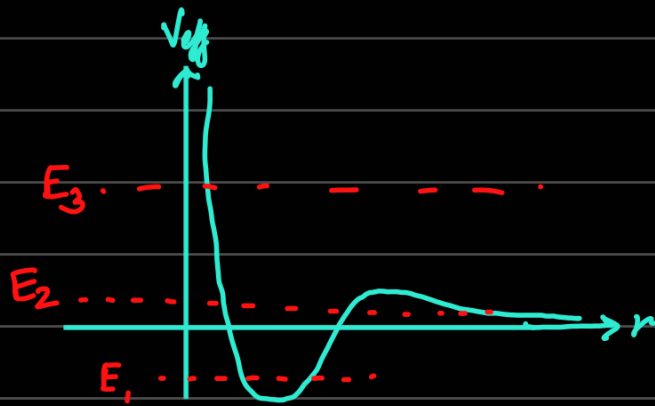
and unstable circular orbit is possible for $E = V_{eff}(ax_0)$



Case 3 : $\frac{2b}{a} < 0.84$

If $E < 0$ like $E = E_1$,
then oscillations possible.

If $E > 0$ and is like $E = E_2$
then oscillations possible.



Circular orbits possible (one stable, and the other unstable)

Unbounded motion for $E > 0$ and something like $E = E_3$.

c) Recall, $\frac{d^2 u}{d\phi^2} + u = -\frac{m}{l^2} \frac{dV}{du}$ $u \equiv 1/r$

let $g(u) = -\frac{m}{l^2} \frac{dV}{du}$

And let u_0 correspond to stable circular orbit.

Then for $u = u_0 + \delta u \rightarrow$ perturbation.

$$g(u) = g(u_0) + g'(u_0) \delta u$$

$$\Rightarrow \frac{d^2}{d\phi^2} (u_0 + \delta u) + (u_0 + \delta u) = g(u_0 + \delta u)$$

Notice how we have assumed $1 - g'(u_0) > 0$. This is because of assumed stability.

$$\Rightarrow \frac{d^2 \delta u}{d\phi^2} + \beta^2 \delta u = 0 \quad \text{with } \beta^2 = 1 - g'(u_0)$$

This corresponds to $\delta u = \alpha \cos \beta(\phi - \phi_0)$

Now let's find $g'(u_0)$, to find β .

Firstly, let's find u_0 , then we will find $g'(u_0)$.

Since r_0 corresponds to circular orbit,

$$\frac{dV_{\text{eff}}}{dr} = 0$$

$$\frac{r^2}{a^2} e^{-\frac{r}{a}} + \frac{r}{a} e^{-\frac{r}{a}} - \frac{2b}{a} = 0$$

or, $\frac{1}{a^2 u^2} e^{-\frac{1}{au}} + \frac{1}{au} e^{-\frac{1}{au}} - \frac{2b}{a} = 0$

$$\Rightarrow 1 + au - 2abu^2 e^{\frac{1}{au}} = 0$$

$$\Rightarrow e^{-\frac{1}{au_0}} = \frac{2abu_0^2}{1 + au_0} \quad - (2)$$

Now $g(u) = -\frac{1}{2A} \frac{dV}{du}$ with $V = -Kue^{-\frac{1}{au}}$

$$\frac{dV}{du} = -K e^{-\frac{1}{au}} - \frac{K}{au} e^{-\frac{1}{au}}$$

$$g(u) = \frac{K}{2A} \left(1 + \frac{1}{au}\right) e^{-\frac{1}{au}}$$

$$g'(u) = \frac{K}{2A} \left(-\frac{1}{au^2}\right) e^{-\frac{1}{au}} + \frac{K}{2A} \left(1 + \frac{1}{au}\right) \frac{e^{-\frac{1}{au}}}{au^2}$$

$$g'(u) = \frac{K e^{-\frac{1}{au}}}{2A a^2 u^3}$$

And from 2,

Recall $\frac{A}{K} = b$

$$g'(u_0) = \frac{K}{2A a^2 u_0^3} \frac{2ab u_0^2}{(1 + a u_0)} = \frac{1}{a u_0 (1 + a u_0)}$$

$$\text{So } \beta^2 = 1 - g'(u_0) = 1 - \frac{1}{a u_0 (1 + a u_0)}$$

$$\Rightarrow \beta = \sqrt{1 - \frac{1}{a u_0 (1 + a u_0)}} \quad ; \text{ Exact rate of precession.}$$

For $\frac{1}{a u_0} \ll 1$,

$$\beta \approx \sqrt{1 - \frac{1}{a^2 u_0^2 (1 + \frac{1}{a u_0})}} = \sqrt{1 - \frac{1}{a^2 u_0^2}} = 1 - \frac{1}{2a^2 u_0^2}$$

And so in one revolution, $\Delta\phi = \frac{2\pi}{\beta}$

$$\begin{aligned} \Rightarrow \Delta\phi &\approx \frac{2\pi}{1 - \frac{1}{2a^2 u_0^2}} = 2\pi \left(1 + \frac{1}{2a^2 u_0^2}\right) \\ &= 2\pi + \frac{\pi}{a^2 u_0^2} \end{aligned}$$

So the orbit precesses by $\boxed{\frac{\pi r_o^2}{a^2}}$ in one revolution.